

Integration by Parts



The integration by parts formula

- 1 Evaluate $\int x e^x dx$ using integration by parts.
- 2 Evaluate $\int x \cos x dx$ using integration by parts.
- 3 Evaluate $\int \ln x dx$ using integration by parts (with $dv = dx$).

Repeated and definite applications

- 4 Evaluate $\int x^2 e^x dx$, applying integration by parts twice.
- 5 Evaluate $\int_0^1 x e^{2x} dx$ exactly.